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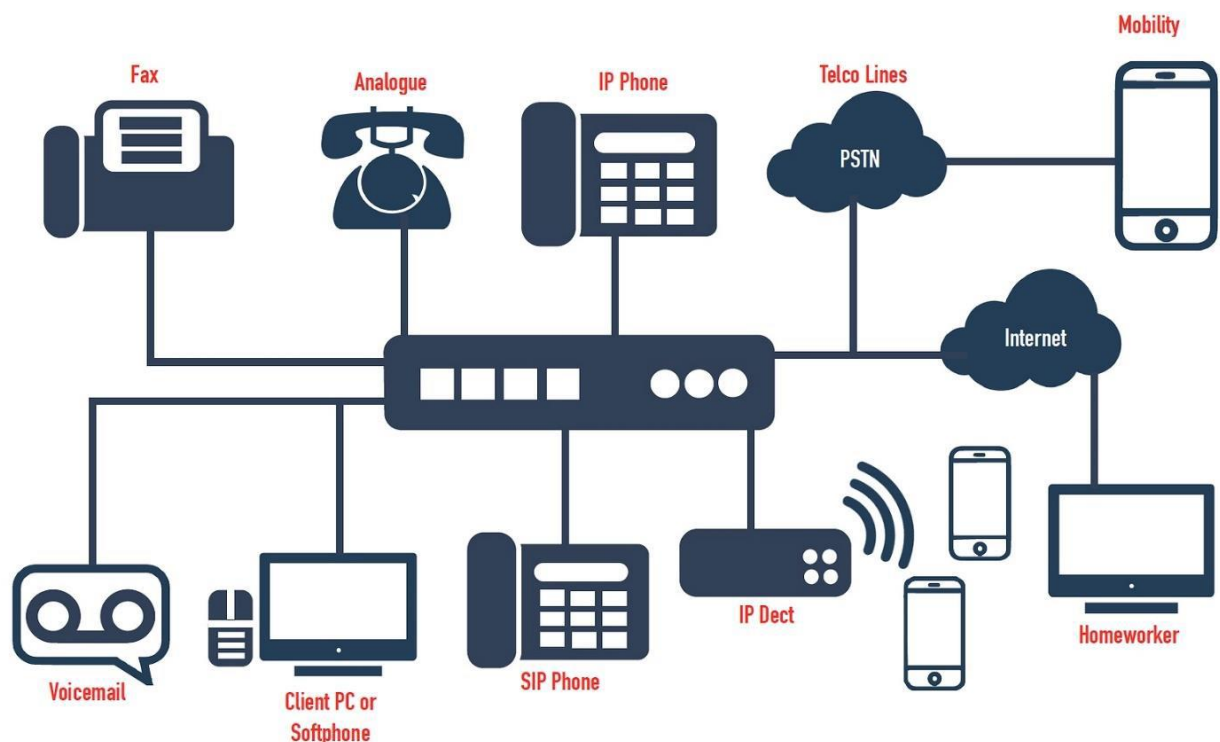
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CO-Mapping

	CO1	CO2	CO3
Q1	✓		
Q2	✓		
Q3	✓		
Q4	✓	✓	

Objective – IP Telephony: Understanding Call Manager, Dynamic Host Configuration Protocol (DHCP), VOICE LAN, Express Telephony Services, and Phone Directory Services.

IOS Telephony Service provides basic IP phone call-handling capabilities in a LAN environment. This service enables the multiservice routers to act as the IOS Telephony Service for IP. This service also helps download phone software images and configures and manages the IP phones in your LAN. When IOS Telephony Service is configured, the IP phones receive initial configuration information and phone firmware from the TFTP server. In most cases, the IP phones obtain the IP address of their TFTP server with the Dynamic Host Configuration Protocol (DHCP) option 150 command.



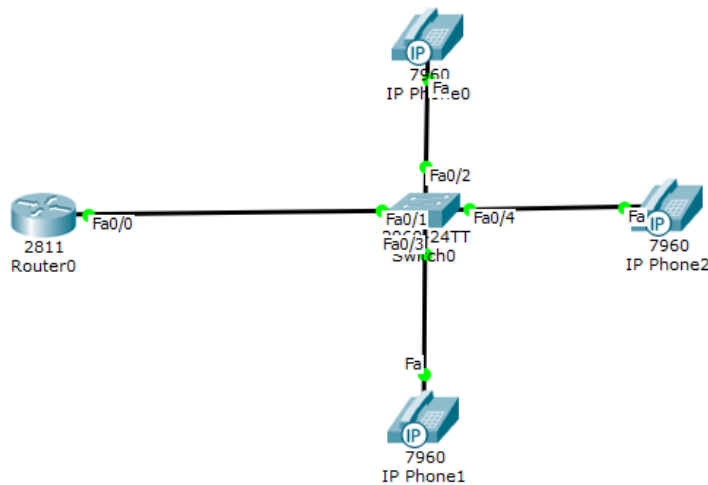
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Provisioning

The router provides a mechanism to provision IOS Telephony Service. This provisioning interface allows you to perform the following functions:

- ❖ Assign directory numbers to the line appearances on each IP phone.
- ❖ Assign numbers to the speed-dial buttons on each IP phone.
- ❖ Assign caller identification information to each directory number.
- ❖ Assign directory numbers to phones other than IP phones attached to the system by using the standard voice-port and dial-peer configuration CLI.
- ❖ Provide dial-plan information to route calls either to public switched telephone network (PSTN) lines or voice network connections.

Task 1: Implement the following topology in Packet Tracer.



Task 2: Router Configuration and Port On.

```
Router>enable
Router#configure terminal
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.1.254 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
```

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Task 3: DHCP Server Configuration.

DHCP server is a full usage server that allocates and oversees IP addresses from determined location pools inside the gadget to DHCP customers. The DHCP server can be arranged to allocate extra parameters, for example, the IP address of the Domain Name System (DNS) server and the default devices. DHCP is an application layer protocol used to distribute various network configuration parameters to devices on a TCP/IP network. – IP addresses, subnet masks, default gateways, DNS servers, etc. DHCP employs a client-server architecture; a DHCP client is configured to request network parameters from a DHCP server on the network. A DHCP server is configured with a pool of available IP addresses and assigns one of them to the DHCP client.

The DHCP server is needed to provide each IP phone connected to the network with an IP address and the TFTP server location.

```
Router(config)#ip dhcp pool VOIPAssignment
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.254
Router(dhcp-config)#option 150 ip 192.168.1.254
```

Task 4: Configuring Express Telephony Services.

Call Manager Express Telephony service is required to enable VOIP on the network.

The max-dn command specifies the maximum number of directory numbers or virtual voice ports supported by the router. The maximum number is IOS and platform dependent. The default number is zero. To see the maximum allowable number of directory numbers that this system can use enter the command max-dn ?

The max-ephones command is used for registering SCCP Phones. The max-ephones command is enhanced to set the maximum number of SCCP phones that can register to Cisco Unified CME, without limiting the number that can be configured. Also, the number of SCCP phones that can be configured is no longer limited to the maximum number that can be registered. Also, it is not limited to the hardware platform. This change expands the maximum number of phones that can be configured to 1000. This enhancement supports features such as Extension Assigner that require you to configure more phones than can register. For example, if you set the max-ephones command to 40 and configure 80 ephones, only 40 phones can register to Cisco Unified CME, one at a time in random order. The remaining 40 phones cannot register, and an error message displays for each rejected phone. This enhancement also allows you to assign ephone tags that match the extension number of the phone, for extensions up to 1000.

The auto assign command allows you to distribute specific ephone-dn ranges to specific types of phones or to any phone requesting an extension. For example, to auto-assign ephone-dns 20 through 24 to Cisco 7940 IP Phones, ephone-dns 25 through 30 to Cisco 7960 IP Phones, and ephone-dns 31 through 39 to any other phone model that attempts to register.

```
Router(config)#telephony-service
Router(config-telephony)#max-dn 10
Router(config-telephony)#max-ephones 10
Router(config-telephony)#ip source-address 192.168.1.254 port 2000
Router(config-telephony)#auto assign 4 to 6
Router(config-telephony)#auto assign 1 to 5.
```

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Task 6: Configuring Switch for Voice LAN.

A voice LAN is a virtual local area network that is specifically allocated for user's voice data streams. It ensures the quality of voice traffic by improving the transmission priority of it when transmitted with other traffic. When other services (data, video, etc.) are transmitted simultaneously, voice service will be prioritized and transmitted with a higher forwarding priority.

```
Switch>enable
Switch#configure terminal
Switch(config)#interface range fastethernet 0/2 – 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport voice vlan 1
```

Task 7: Configuring Phone Directory.

```
Router>enable
Router#config t
Router(config)#ephone-dn 1
Router(config-ephone-dn)#number 12345
Router(config-ephone-dn)#ephone-dn 2
Router(config-ephone-dn)#number 12354
Router(config-ephone-dn)#ephone-dn 3
Router(config-ephone-dn)#number 13245
```

Task 8: Verify the Configuration.

Ring IP Phone 0 and IP Phone 2 from IP Phone 1. If the connection was successful, the beep can be heard, or a red light is triggered on GUI.